

ABUMUKIZA LEON CHRISTIAN

Software Engineer (Java) & AI/Machine Learning Engineer (Python)

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[**\[Github\]**](#)

[**\[Linked in\]**](#)

PROFESSIONAL SUMMARY

Results-driven Software Engineering student specializing in scalable backend systems with Java and intelligent data solutions with Python. Experienced in building full-cycle applications from architecture to deployment, with a strong command of Object-Oriented Design, Data Structures, and Algorithms. Passionate about leveraging AI and Machine Learning to solve high-impact problems. Actively preparing for the Africa to Silicon Valley (A2SV) program with a mission to build world-class technology infrastructure rooted in Africa.

KEY PROJECTS

AgriStock – Agricultural Inventory Management System | Java, MySQL

- Architected and developed a full-stack inventory management platform tailored for Rwandan agribusinesses to track crop yields, stock levels, and supply chain logistics in real time.
- Implemented robust OOP principles including inheritance, polymorphism, and encapsulation to create a modular, scalable codebase that supports multiple user roles (farmers, distributors, admins).
- Designed and optimized relational database schemas with complex SQL queries to handle large volumes of agricultural data, reducing stock discrepancy errors and improving inventory accuracy.

XtonFlow – Workflow & Transaction Processing Engine | Java, Spring Boot

- Built a backend workflow automation engine designed to streamline multi-step business processes and transaction pipelines for small-to-medium enterprises.
- Developed RESTful API endpoints with Spring Boot to handle secure data exchange between client interfaces and the core processing engine, ensuring low-latency response times.
- Integrated advanced Java concurrency and exception-handling mechanisms to maintain system stability under high transaction loads, demonstrating production-grade software engineering practices.

Fraud Detection System – ML-Powered Anomaly Classifier | Python, Scikit-Learn, Pandas

- Engineered an end-to-end machine learning pipeline to detect fraudulent financial transactions by analyzing patterns in large-scale transactional datasets.
 - Performed extensive data preprocessing including handling missing values, feature scaling, class balancing (SMOTE), and feature engineering to maximize model performance on imbalanced data.
 - Trained and evaluated multiple classification models (Random Forest, Logistic Regression, Gradient Boosting), achieving high precision and recall metrics, and deployed the best-performing model for real-time fraud flagging.
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EDUCATION

bachelor's degree in IT - Software Engineering & Information Management

Adventist University Of Central Africa

2023 - Present

A2 - High School Diploma (Mathematics - Economics - Computer Science)

Ecole Secondaire Mudasomwa

2020 - 2022

Advance Level (Mathematics - Economics - Computer Science)

Ecole Secondaire Marie Adelaide

2019 - 2020

Ordinary Level (General Education)

Petit Seminaire Virgo Fidelis De Butare

2016 - 2018

ADDITIONAL INFORMATION

TECHNICAL SKILLS

- Languages: Java (Advanced), Python (Advanced – AI/ML), SQL
- Java Ecosystem: OOP, Collections Framework, Multithreading, Spring Boot, JDBC, Maven
- AI/ML Stack: Scikit-Learn, Pandas, NumPy, Matplotlib, TensorFlow (Fundamentals)
- Core Competencies: Data Structures & Algorithms, System Design, Database Management, RESTful API Design, Data Preprocessing & Feature Engineering
- Tools: Git/GitHub, IntelliJ IDEA, PyCharm, VS Code, MySQL, PostgreSQL
- Digital & Creative: Digital Marketing Strategy, Video Editing (Adobe Premiere Pro, DaVinci Resolve), Graphic Design (Adobe Photoshop, Illustrator, Figma), Content Creation, Brand Identity.

LANGUAGE

- English
- French
- Kinyarwanda

Reference

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